

TRANS3 SMART ADMIN INTERFACE V3.1

Layer 3 In-Place Payload Encryptor — Operational Reference Guide

1. Introduction

TRANS3 is an IP packet encryption engine operating at the network layer (L3). Unlike traditional VPNs, it encrypts only the payload without modifying IP or transport headers. TCP, UDP, ICMP packets retain their nature on the wire; only the actual data becomes indecipherable.

Technical specifications:

- **XChaCha20-Poly1305** authenticated encryption (256-bit key, 192-bit nonce)
- No session negotiation (zero latency)
- Fixed **32 bytes** overhead per packet
- Anti-replay protection via 4096-packet sliding window
- Proprietary AAD to reject any non-TRANS3 traffic

2. First launch: interface selection

Upon startup, the script automatically detects physical network interfaces and asks you to select the one you want to protect.

```
Select the interface to protect with TRANS3

1. enp0s31f6
2. wlp2s0

Your choice (1-2) : 2
```

Figure 1: Interface selection at startup

Once the interface is chosen, the engine is initialized and the main menu appears.

3. Main menu

```

TRANS3 SMART IP ENCRYPTOR V3.1 (L3)

Protected Interface : wlp2s0
Kernel Engine      : Operational [V]
Active Target IPs  : 0

ENCRYPTION MATRIX (POSTROUTING/PREROUTING) :
> Standard traffic (No encryption rules)

1. Initialize kernel engine
2. Security Key Management
3. Add an encryption rule (Target IP)
4. Real-time Encryption Telemetry
5. Lock system configuration (ROM write)

```

```
6. Purge matrix (Clear Text Mode)
7. Exit

Command (1-7) :
```

Figure 2: Main menu (no active rules)

The menu displays in real-time the protected interface, kernel module status, number of active targets, and the encryption matrix.

4. Detailed options

4.1 Option 1 – Initialize kernel engine

Re-initializes the engine: reloads the xt_TRANS3 module, disables hardware offloads, and adjusts the MTU to 1550 to absorb the 32-byte overhead.

```
[*] Re-initializing kernel engine...
[+] Engine restarted successfully. MTU adjusted to 1550 on wlp2s0.

Press Enter to continue...
```

Figure 3: Engine initialization

4.2 Option 2 – Security Key Management

Allows you to check the current key or import a new one from a removable medium (air-gap).

```
KEY SECURITY

a) Check local signature status
b) Import external token (Air-Gap USB)

[+] Active key :
d32c0aa45615a2e6c5584b764c9b92317fd3ce7ae45bbe3e85604e444f8459aa036429c768a478fb31e3244ff8f6063ff4323186e7

Press Enter to continue...
```

Figure 4: Displaying the active key

4.3 Option 3 – Add an encryption rule

Adds an encryption rule for a target IP. All protocols (TCP, UDP, ICMP, IGMP, GRE) are automatically covered.

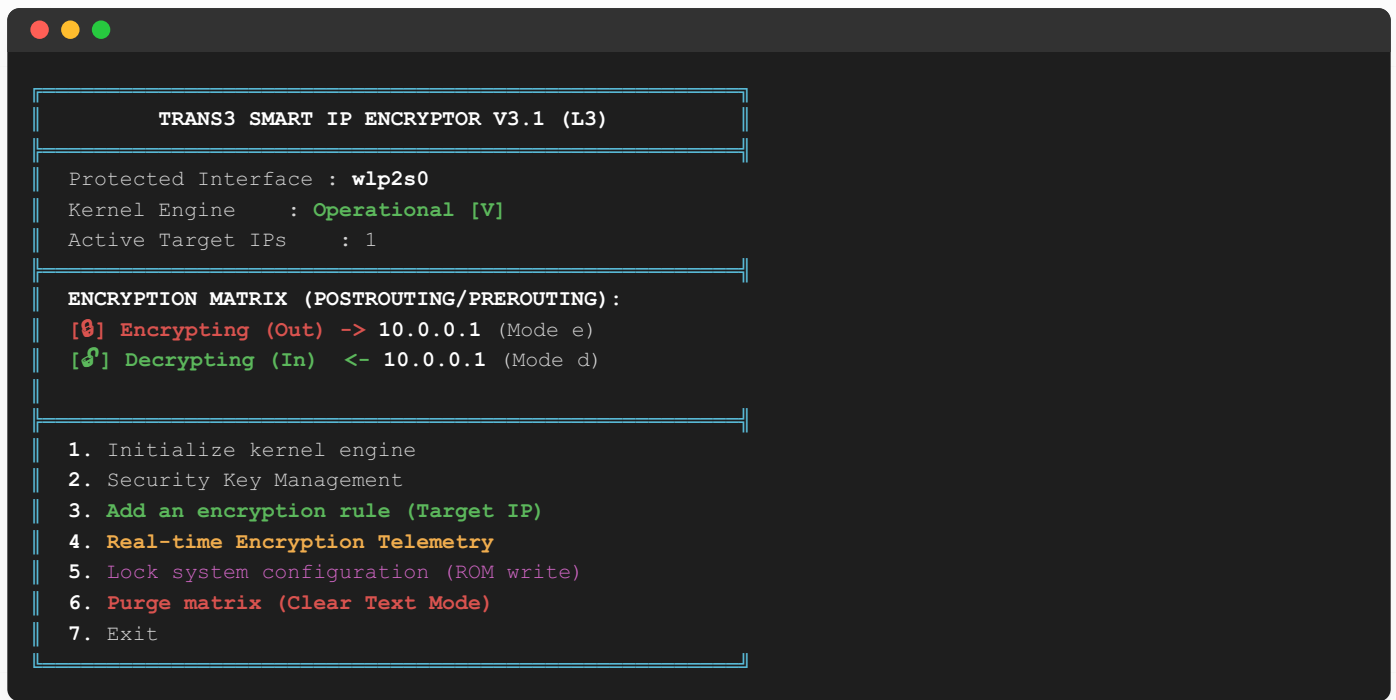
```
ADD L3 ENCRYPTION RULE

10.0.0.1
[+] Rules added. Traffic to/from 10.0.0.1 is now encrypted (Layer 3).

Press Enter to continue...
```

Figure 5: Adding an encryption rule

After addition, the main menu updates:



```
TRANS3 SMART IP ENCRYPTOR V3.1 (L3)

Protected Interface : wlp2s0
Kernel Engine      : Operational [V]
Active Target IPs  : 1

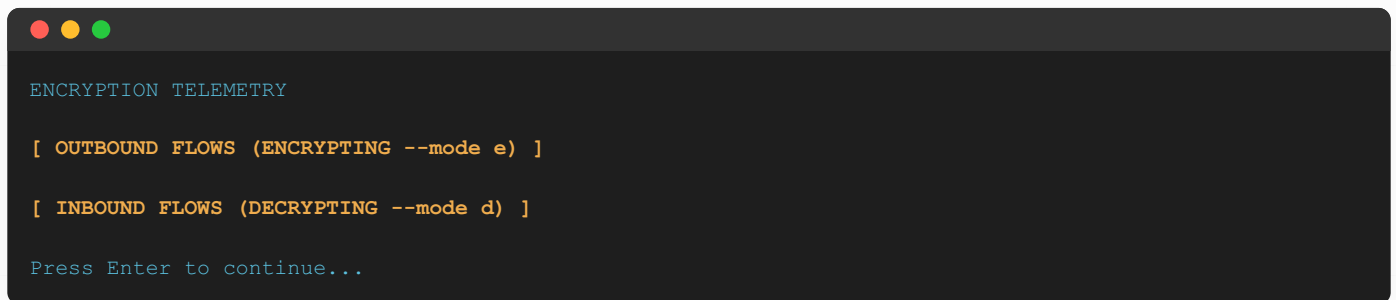
ENCRYPTION MATRIX (POSTROUTING/PREROUTING) :
[0] Encrypting (Out) -> 10.0.0.1 (Mode e)
[0] Decrypting (In)  <- 10.0.0.1 (Mode d)

1. Initialize kernel engine
2. Security Key Management
3. Add an encryption rule (Target IP)
4. Real-time Encryption Telemetry
5. Lock system configuration (ROM write)
6. Purge matrix (Clear Text Mode)
7. Exit
```

Figure 6: Menu with one active rule

4.4 Option 4 – Real-time Encryption Telemetry

Displays packet counters for encrypted and decrypted flows per target.



```
ENCRYPTION TELEMETRY

[ OUTBOUND FLOWS (ENCRYPTING --mode e) ]

[ INBOUND FLOWS (DECRYPTING --mode d) ]

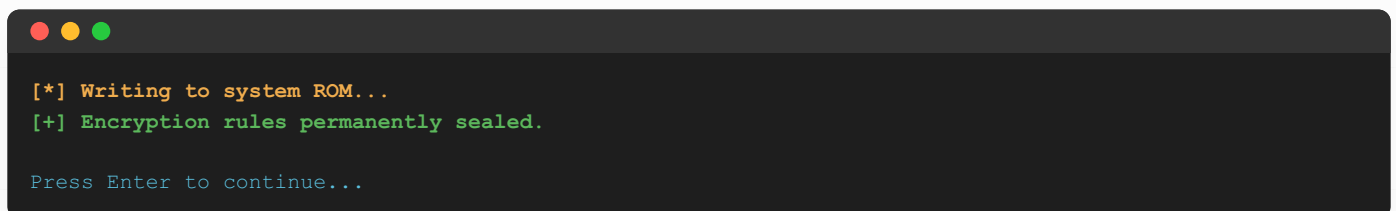
Press Enter to continue...
```

Figure 7: Telemetry (no traffic yet)

Note: Counters will appear after traffic has been exchanged.

4.5 Option 5 – Lock system configuration

Saves iptables rules to /etc/iptables/rules.v4 to make them persistent across reboots.



```
[*] Writing to system ROM...
[+] Encryption rules permanently sealed.

Press Enter to continue...
```

Figure 8: Persistence of rules

4.6 Option 6 – Purge matrix

Removes all encryption rules and restores standard network configuration (MTU 1500, offloads re-enabled).



```
[*] Purging encryption rules...
[+] Network interface restored to Clear Text Mode (MTU 1500).

Press Enter to continue...
```

Figure 9: Purge of rules

4.7 Option 7 – Exit

Exits the interface while leaving the engine active in the background (or disabling it, according to your choice).

```
[?] Shutdown sequence:
Keep encryption active in background? (Y/n): Y
> Encryption active in background.
Terminal disconnected.
```

Figure 10: Exit with engine kept running

5. Conclusion

TRANS3 provides a lightweight, transparent, and secure network encryption solution. Thanks to its intuitive administration interface, deploying confidentiality on critical links becomes effortless. For any questions, consult the technical documentation or contact your administrator.